1) Explain the difference between these two statements

A) "The loss function defines the landscape"
B) "Gradient descent defines the path through the landscape"

Then identify which object each phrase refers to in the update rule

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

What we mean by A is that the loss function scores a particular set of parameters (θ_t) with how bad it is.

By B we mean that for any point θ_t, we find the gradient (how much our loss function changes based on a perturbation of θ_t, and in what direction is the highest change). This tells us which direction is down hill, and how steep it is. Then our new position θ_{t+1} is our current position adjusted by step size η and the negative of the vector pointing downhill.

The gradient sort of acts like a velocity, and η like a time step for our analogy of a ball rolling downhill (with no momentum)

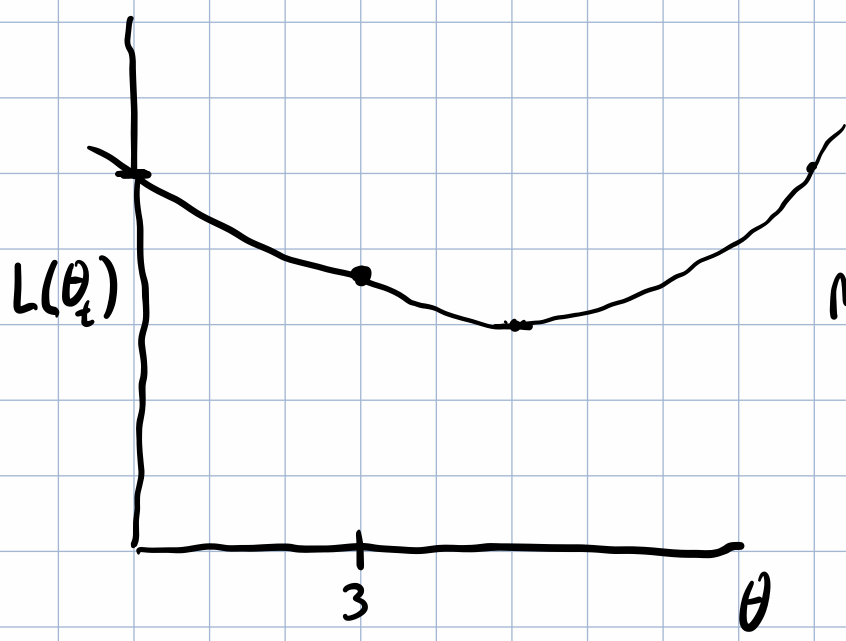
2) Draw a one-dimensional loss curve with $\theta = 3$. Mark the current point $\theta_t < 3$, then label

A) The sign of the gradient
B) The downhill direction
C) A small learning rate step
D) A large learning rate step that might over shoot the minimum

Suppose

$L(\theta_t) = \frac{1}{5}(\theta - 5)^2 + 3$

$\nabla L(\theta_t) = \frac{d}{d\theta} L(\theta_t)$
$= \frac{2}{5}(\theta - 5)$
$= \frac{2}{5}\theta - 2$



Minimum at $\nabla L(\theta_t) = 0$
$0 = \frac{2}{5}\theta - 2$
$2 = \frac{2}{5}\theta$
$5 = \theta$

Evaluate at $\theta_0 = 3$

$L(\theta_0) = \frac{1}{5}(-2)^2 + 3$
$= \frac{4}{5} + 3$

$\nabla L(\theta_0) = \frac{6}{5} - 2 = -\frac{4}{5}$

$\text{sign}(\nabla L(\theta_0)) = -1$ A downhill direction

$-\nabla L(\theta_t) = 4/5$ B

A small learning rate step $\eta_{small} = 1/5$ C

A problematically large step $\eta_{large} = 5$ D

$$\theta_{1_{small}} = \theta_0 - \eta_{small} \nabla L(\theta_0)$$

$$= 3 - \frac{1}{5} \cdot \frac{4}{5} = 11/5$$

$$\theta_{1_{large}} = \theta_0 - \eta_{large} \nabla L(\theta_0)$$

$$= 3 - 5\,(-4/5)$$

$$= 3 + 4 = 7 \quad \text{(overshot minima @ } \theta = 5\text{)}$$

3) Let

$$L(\theta) = (\theta - 4)^2$$

then

$$\nabla L(\theta) = \tfrac{d}{d\theta} L(\theta)$$
$$= 2(\theta - 4)$$
$$= 2\theta - 8$$

Start with

$\theta_0 = 1 \qquad \eta = 0.1$

Compute $\theta_1, \theta_2, \theta_3$

for each step write the gradient value before applying the update

$$\nabla L(\theta_0) = 2\theta_0 - 8 = -6$$

Too lazy for a calculator

$$\theta_1 = \theta_0 - \eta \nabla L(\theta_0) = 1 - \tfrac{1}{10}(-6) = \tfrac{8}{5} = 1.6$$
$$\nabla L(\theta_1) = 2(\tfrac{8}{5}) - 8 = \tfrac{16}{5} - 8 = \frac{16-40}{5} = -\tfrac{34}{5}$$

$$\theta_2 = \theta_1 - \eta(\nabla L(\theta_1)) = \tfrac{8}{5} - \tfrac{1}{10}(-\tfrac{34}{5}) = \frac{80+34}{50} = \frac{114}{50} \approx 2.3$$

$$\begin{array}{r} {}^{2} \\ 114 \\ 5 \\ \hline 570 \end{array}$$

$$\nabla L(\theta_2) = 2\left(\tfrac{114}{50}\right) - 8 = \frac{114}{25} - \frac{200}{25} = -\frac{86}{25}$$

$$\theta_3 = \frac{114}{50} - \frac{1}{10}\left(\frac{-86}{25}\right) = \frac{114}{50} + \frac{86}{250} = \frac{570+86}{250} = \frac{656}{250}$$

$$= \frac{328}{125} \approx 2.6$$

$$\begin{array}{r} 2.6 \\ 125\overline{)328.0} \\ 250 \\ \hline 780 \\ 725 \\ \hline 55 \end{array}$$

$\theta_0 = 1$
$\theta_1 \approx 1.6$
$\theta_2 \approx 2.3$
$\theta_3 \approx 2.6$

4) Suppose

$$L'(\theta) = 10\,L(\theta)$$

4.1 Does the minimizer of L' change compared to the minimizer of L?
4.2 What happens to the gradient?
4.3 If the origional learning rate was η, what learning rate would make gradient descent take the same steps on L'?

4.1 If $\eta = \eta'$, then the minimizer of L' would use a step size 10 times that of L

4.2 $\nabla L' = 10 \nabla L$

4.3 In order for L to take the same number of steps as L',
$\eta' = 1/10\,\eta$

$$\theta_1 = \theta_0 - \eta \nabla L$$

$$\theta_1' = \theta_0 - \eta' \nabla L' = \theta_0 - \eta'(10 \nabla L)$$

$$\theta_1' = \theta_1$$

$$\cancel{\theta_0} + \eta \cancel{\nabla L} = \cancel{\theta_0} + 10\eta' \cancel{\nabla L}$$

$$\eta' = 1/10\,\eta$$

5) Let

$$L(w,b) = (w-1)^2 + 9(b+2)^2$$

The gradient is

$$\nabla L(w,b) = \begin{bmatrix} 2(w-1) \\ 18(b+2) \end{bmatrix} = \begin{bmatrix} \partial/\partial w\, L \\ \partial/\partial b\, L \end{bmatrix}$$

Start at

$$(w_0, b_0) = (0,0) \qquad \eta = 0.05$$

Compute one gradient descent update. Then explain why the b-coordinate moves more dramatically than the w-coordinant

$\theta_0 = (w_0, b_0) = (0,0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\theta_1 = \theta_0 - \eta \nabla L_0 = 0 - 0.05 \begin{bmatrix} 2(-1) \\ 18(2) \end{bmatrix} = \begin{bmatrix} 0.1 \\ 1.8 \end{bmatrix}$

$\partial/\partial b \, L = 18(b+2) \quad \partial^2/\partial b^2 \, L \quad 18$

$\partial/\partial w \, L \quad 2(w-1) \quad \partial^2/\partial w^2 \, L = 2$

The b-coordinate varies 9 times faster. (It's right there in the coefficients)

6) Use the linear approximation

$L(\theta + \Delta\theta) \approx L(\theta) + \nabla L(\theta)^T \Delta\theta$

Show what happens when

$\Delta\theta = -\eta \nabla L(\theta) \quad$ for $\eta > 0$

Explain why the dot product term is non-positive

$L(\theta - \eta \nabla L(\theta)) \approx L(\theta) + \nabla L(\theta)^T \cdot (-\eta \nabla L(\theta))$

$L(\theta_{t+1}) \approx L \overset{-}{-} \overset{+}{\eta} \overset{+}{\nabla L^T \cdot \nabla L}$

$\nabla L^T \cdot \nabla L$ will always be either both positive or both negative, so their product is positive. η is defined to be positive, so we're left with the negative from the definition of $\Delta\theta$.

7) A regularized objective has the form

$$J(\theta) = L_{data}(\theta) + \lambda \|\theta\|_2^2$$

A) Which part measures data fit
B) Which part penalizes large parameters
C) Which object would gradient descent use to update θ: $\nabla L_{data}(\theta)$ or $\nabla J(\theta)$
D) How is changing λ different from changing the learning rate η

A) λ

B) $\|\theta\|_2^2$

loss = prediction error + λ * penalty

C) $J(\theta)$

D) Changing λ stretches the gradient of the loss landscape; Changing the learning rate changes how "fast" we go down a loss landscape relative to its gradient.

Implementation problems in `exercises/lesson-26-

gradient-descent/implementation.ts`

Reflection

- Which part still feels symbolic rather than geometric?

  I think I have a good grasp on how to think about gradient descent.

- Did the 1-D or 2-D example make the idea clearer?

  I came into this with a lot of background knowledge, so I'm not sure.

- What is the clearest difference between regulation strength λ versus learning rate η?

  λ adjusts the landscape
  η adjusts the path through it

- What should the next lesson reinforce before stochastic gradient descent adds mini-batches?